

Rachit Tamrakar

LinkedIn | Portfolio

U.S. Lawful Permanent Resident (ITAR eligible)

tamrakar.rachit2004@gmail.com

+1 (425) 574-2708

Redmond, WA

WORK EXPERIENCE

Bosch Australia Manufacturing Solutions (BAMS) Undergraduate Vision, Robotics and Simulations Engineer

Melbourne, Australia
Mar 2024 - Present

Vision Inspection System

- Designed and deployed a 6-station automated vision inspection system on an **\$8M** production line, integrating 40+ cameras, **HALCON** vision library, and **SQL** traceability into a multithreaded **C#/.NET** application.
- Architected an asynchronous pipeline separating UI, PLC communications (**PROFINET**), and image acquisition, achieving sub-second cycle times and serving as a template for future department projects.
- Resolved critical integration issues during commissioning, including debugging race conditions in the threading model and optimizing network configurations for high-bandwidth traffic.

Real-Time Object Detection System (YOLO)

- Built a real-time detection pipeline for product positioning, training a compact **YOLO** model (**PyTorch**) to achieve 300 ms inference latency on legacy hardware (1.5x faster than previous baselines).
- Developed automated dataset tooling for image collection, auto-labeling, and model training to accelerate the deployment of deep learning models on the factory floor.

Digital Twin Simulation

- Built high-fidelity **NX** Mechatronics Concept Designer simulations and **Python** automation scripts to enable Software-in-the-Loop (SIL) virtual commissioning of **PLC** logic.
- Automated the mapping of hundreds of I/O points, reducing test preparation time by 2x and allowing validation of robotic palletizer features without halting production.

PROJECTS / EXTRACURRICULARS

Monash High Powered Rocketry (Monash HPR) Avionics Vice Lead – Propulsion

Melbourne, Australia
Aug 2023 - Present

Solaris Mk II Hybrid Engine & Ground Station Electronics (IREC 2025)

- Designed and built the ground station electrical cabinet and distributed control system for a hybrid rocket engine (Raspberry Pi, LabJack T7, STM32), implementing Python state machines, safety interlocks, and MQTT telemetry.
- Owned test readiness for 8+ hot-fire campaigns and served as console operator for 4 live tests, issuing go/no-go status based on sensor sanity and telemetry health.
- Co-presented the control architecture at the 2025 International Rocketry Engineering Competition (IREC), contributing to the team winning the Jim Furfaro Award for Technical Excellence (140+ teams) and placing 2nd in category.

Capacitive Propellant Sensor

- Led a 4-person team to develop a capacitive **N₂O** mass sensor, co-designed the probe geometry, wrote **STM32** firmware (**C**) for capacitance measurement and signal processing and ran calibration/verification procedures.
- Debugged and mitigated parasitic-capacitance noise via shielding (actively-driven shielded cable), achieving **±10 g** repeatability, enabling reliable autonomous fill control.

EDUCATION

Monash University Bachelor of Engineering - Mechatronics (Honours)

Melbourne, Australia
Feb 2022 - Dec 2026

High Distinction (83.714 WAM, 3.64 GPA) | Deans Honours List (2022, 2023)

SKILLS

General: Systems Engineering, Real-Time Systems, DAQ & Instrumentation Systems, 3D printing, Networking

Programming: Git, C/C++, C#/.NET, WPF, Python, SQL, InfluxDB, PyTorch, OpenCV, HALCON, MATLAB/Simulink

Software / Robotics: Multithreaded Programming, State Machines, Embedded Firmware, PROFINET, Communication Protocols (TCP/UDP, MQTT, RS-422, CAN), YOLO, Algorithms & OOP, SLAM, PID Control, Signal Processing

Automation & Design: Siemens PLC, Ignition SCADA, Industrial Cameras, LabJack T7, NX Mechatronics Concept Designer, Solidworks, Altium Designer, STM32CubeIDE

Electronics & Platforms: Linux, Raspberry Pi, STM32, Arduino, Oscilloscope & DMM, Circuit Design & Debugging, Soldering